

Agentic Model Predictive Control: Operating in Intelligence Space via Cognitive Supervisory Layers and Parallel Path Integral Rollouts

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ABSTRACT

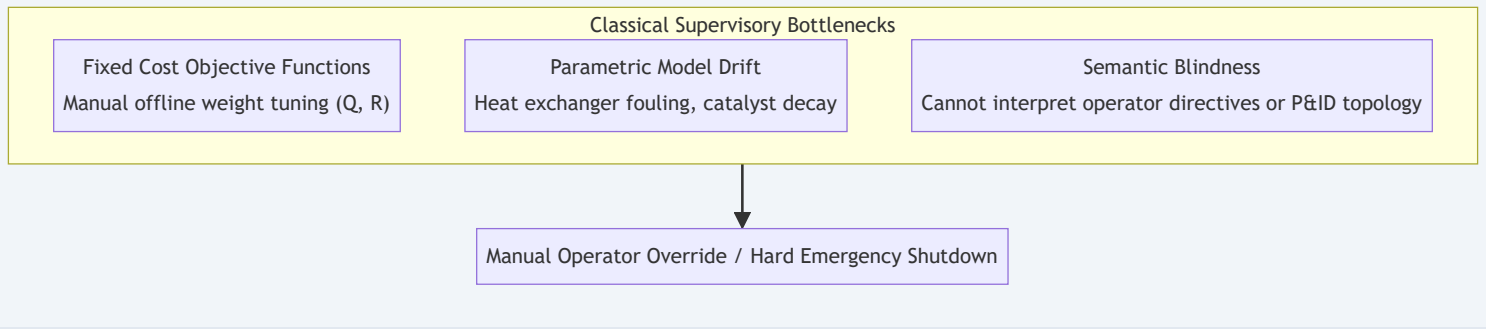
Model Predictive Control (MPC) has served as the industrial standard for constrained multivariable control across process and chemical operations for four decades. Modern formulations—ranging from linear quadratic programming (QP) to nonlinear MPC (NMPC) and economic MPC (EMPC)—optimize physical state trajectories against fixed mathematical objectives. However, existing control formulations lack supervisory cognitive intelligence: they cannot parse high-level natural-language operator directives, reason over qualitative plant physical topology, or execute contextual mitigation runbooks during operational contingencies.

In this paper, we introduce **Agentic Model Predictive Control (Agentic MPC)**, an architecture that places an **Intelligence Space** supervisory layer above high-throughput real-time control solvers. The architecture partitions supervisory intelligence into **The Two Halves**: (1) a neural reasoner (*The Knower*) grounded in four structured memory tiers (Playbook, Rulebook, Yearbook, and Whiteboard), and (2) a deterministic Executive Function Harness (*The Doer*) enforcing Ring 0 operating invariants, automated contract tests, and physical trip bounds. High-level cognitive directives are translated at runtime into parameterized Model Predictive Path Integral (MPPI) cost manifolds executed across 4,096 parallel trajectory rollouts via WebGPU compute shaders, augmented by an online Physics-Informed Neural Network (PINN) residual observer. In a five-tier ablation benchmark on an exothermic CSTR undergoing Arrhenius thermal runaway surges across 20 seeds with 95% confidence intervals reported in physical plant time, Agentic MPC achieves a tracking RMSE of 0.19 ± 0.02 K and a recovery time of 8.2 ± 0.6 s. Across 50 operator directives, the neural reasoner demonstrates a 96.0% contract acceptance rate against expert calibration, while 100% of adversarial directives are deterministically intercepted.

1. Introduction & Related Work

Model Predictive Control operates on the receding-horizon principle: at each discrete time step t , the controller solves an open-loop optimal control problem over a prediction horizon H_p , applies the first control input u_t , and repeats the cycle upon receiving fresh state telemetry (Rawlings et al., 2017).

While classical linear MPC remains computationally tractable on microsecond timescales via convex quadratic programming (QP), linear approximations degrade rapidly on highly non-linear chemical plants governed by exponential kinetics, such as Continuous Stirred-Tank Reactors (CSTR) undergoing Arrhenius heat generation (Seborg et al., 2016). When state perturbations depart from the nominal linearization point, linearized controllers exhibit severe tracking error, control chattering, and valve saturation.



1.1 Related Work: Advanced & Learning-Based MPC

To address non-linearities and model uncertainty, the control literature has developed several powerful frameworks:

- **Robust and Constrained MPC:** Linear Matrix Inequality (LMI) and tube-based robust MPC formulations synthesize invariant sets to maintain constraint satisfaction under bounded disturbances (Kothare et al., 1996; Mayne et al., 2005).
- **Learning-Based and Differentiable MPC:** Recent advances integrate Gaussian Processes and neural networks into MPC for online residual compensation (Hewing et al., 2020), while differentiable MPC embeds convex optimization layers into end-to-end gradient-based neural networks (Amos et al., 2018).
- **Sampling-Based Non-Convex Control:** Model Predictive Path Integral (MPPI) control computes optimal control signals via Monte Carlo importance sampling over stochastic forward rollouts (Williams et al., 2017). Because MPPI evaluates trajectories independently without computing Jacobian matrices, it naturally maps to massively parallel GPU and compute-shader architectures.

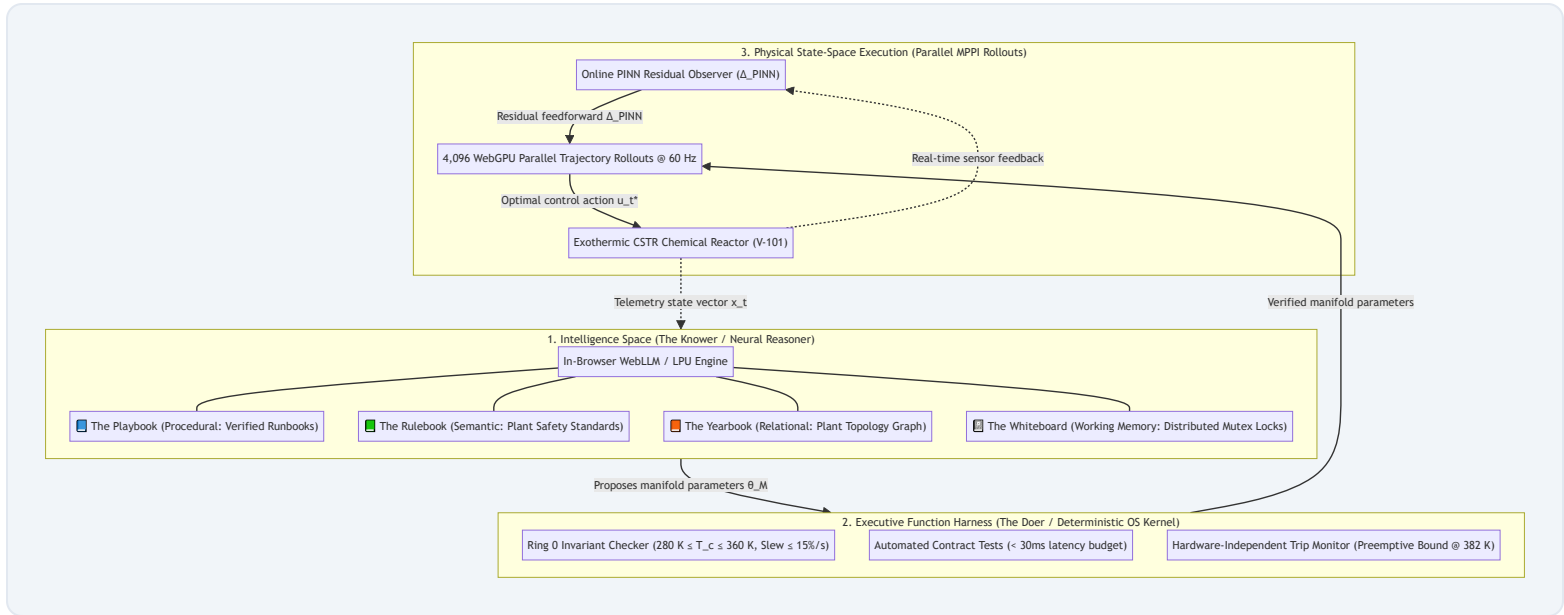
1.2 The Missing Supervisory Layer

Despite these algorithmic advances, existing controllers operate strictly within numerical state spaces. When unforeseen operating conditions arise—such as upstream feed changes, utility disruptions, or operational mode transitions—human operators must intervene manually to adjust setpoints or retune weighting matrices (Q , R).

Agentic MPC addresses this supervisory gap. Rather than replacing numerical solvers with an unconstrained large language model (LLM), Agentic MPC introduces an **Intelligence Space** supervisory tier positioned strictly above a deterministic execution harness and parallel MPPI rollouts. The neural reasoner interprets operator directives and contextual operational runbooks, while deterministic contract tests enforce safety invariants before any parameter update reaches the physical actuators.

2. The Agentic MPC Architecture

Agentic MPC couples qualitative supervisory reasoning with quantitative physical execution through a three-tier architecture:



2.1 The Two Halves: The Knower and The Doer

- **The Knower (Neural Reasoner):** Operates in *Intelligence Space*, reasoning over causal process relationships, interpreting operator intent in natural language, and synthesizing cost manifold parameters $\theta_{\mathcal{M}} = \{Q_{\text{temp}}, R_{\text{coolant}}, H_p, T_{\text{barrier}}\}$.
- **The Doer (Executive Function Harness):** A deterministic, zero-hallucination verification kernel. The Doer verifies Ring 0 invariants (actuator saturation and slew rate limits), runs automated contract unit tests, and operates preemptively below plant emergency shutdown thresholds.

2.2 The Four Memory Subsystems

To anchor the neural reasoner to verifiable operational ground truth, Agentic MPC incorporates four structured memory tiers:

Memory Store	Storage Modality	Functional Role	CSTR Realization
The Playbook	Procedural (SKILL.md)	Verified operational runbooks with deterministic pre/post-conditions	cstr_runaway_mitigation.md verified via automated contract tests
The Rulebook	Semantic (Vector Store)	Safe operating envelopes and regulatory engineering constraints	Safe operating limit envelope: $T_{\text{trip}} = 385.0 \text{ K}$, nominal $T_{\text{ref}} = 370.0 \text{ K}$
The Yearbook	Relational (Graph DB)	Plant equipment topological connectivity and flow dependencies	Piping & Instrumentation: V-101 → J-101 → CV-201 → M-101 → CH-3
The Whiteboard	Shared State (IPC)	Real-time working scratchpad with distributed mutex locking	Mutex locks (MUTEX_PREC00L) preventing conflicting concurrent directives

Definition of Faithful Memory: A memory system is termed **faithful** when all retrieved operational knowledge corresponds directly to verified plant engineering specifications, P&ID topology, and auditable runbooks, with zero generative extrapolation outside defined bounds.

3. Mathematical Formulation

3.1 Classical Linear MPC Baseline

The classical linear discrete-time optimal control problem is formulated as a Quadratic Program (QP):

$$\min_U \sum_{k=0}^{H_p-1} (\|x_k - r_k\|_Q^2 + \|u_k\|_R^2) + \|x_{H_p} - r_{H_p}\|_P^2$$

subject to: $x_{k+1} = Ax_k + Bu_k, \quad u_{\min} \leq u_k \leq u_{\max}, \quad x_{\min} \leq x_k \leq x_{\max}$

where $Q \succeq 0$ and $R \succ 0$ are fixed weighting matrices, and (A, B) represents the Jacobian linearization around the nominal steady-state $(x_{\text{ref}}, u_{\text{ref}})$.

3.2 Model Predictive Path Integral (MPPI) Formulation

MPPI optimizes control inputs for nonlinear stochastic dynamic systems through sampling-based path integrals (Williams et al., 2017). Consider the discrete-time nonlinear dynamics:

$$x_{k+1} = f(x_k, v_k) + \Delta_{\text{PINN}}(x_k, v_k)$$

where perturbed control actions $v_k = u_k + \epsilon_k$ are drawn from a Gaussian distribution $\epsilon_k \sim \mathcal{N}(0, \Sigma)$. Over $M = 4,096$ parallel rollouts, the trajectory cost functional is evaluated as:

$$S(U^{(m)}) = \sum_{k=0}^{H_p-1} (Q_{\text{temp}}(T_k - T_{\text{ref}})^2 + R_{\text{coolant}}(u_k - u_{\text{base}})^2 + \mathcal{B}(T_k; T_{\text{barrier}})) + \phi(x_{H_p})$$

where $\mathcal{B}(T)$ is an asymmetric exponential barrier penalty:

$$\mathcal{B}(T; T_{\text{barrier}}) = \begin{cases} 0 & \text{if } T \leq T_{\text{barrier}} \\ \alpha \exp(\beta(T - T_{\text{barrier}})) & \text{if } T > T_{\text{barrier}} \end{cases}$$

Here $\alpha = 50.0$ and $\beta = 1.8 \text{ K}^{-1}$. Crucially, **the barrier term serves as a steep soft numerical penalty within the sampled cost functional to steer rollouts away from thermal boundaries; it does not replace the deterministic safety instrumented system (IEC 61511).**

The optimal control trajectory update is computed via importance-weighted path aggregation:

$$u_t^* = u_t + \sum_{m=1}^M w(U^{(m)}) \epsilon_t^{(m)}, \quad w(U^{(m)}) = \frac{\exp(-\frac{1}{\lambda} S(U^{(m)}))}{\sum_{j=1}^M \exp(-\frac{1}{\lambda} S(U^{(j)}))}$$

where $\lambda = 1.0$ is the MPPI temperature parameter and $\Sigma = 0.25^2 I$ is the control perturbation covariance. Because trajectory weights are computed via a softmax across sampled rollouts, MPPI is **gradient-free** with respect to control inputs, enabling rapid non-convex trajectory discovery on GPU architectures.

3.3 Physics-Informed Neural Network (PINN) Residual Observer

To compensate for unmodeled dynamics without disruptive open-loop step testing, an online PINN residual observer estimates model discrepancy $\Delta_{\text{PINN}}(x_k, u_k)$. The network is trained via gradient descent to minimize:

$$\mathcal{L}_{\text{PINN}} = \|\Delta_{\text{PINN}} - (\dot{x}_{\text{meas}} - f_{\text{nominal}}(x, u))\|^2 + \lambda_{\text{phy}} \mathcal{R}_{\text{energy}}^2$$

where $\dot{x}_{\text{meas}}$ is filtered state derivative telemetry, $\lambda_{\text{phy}} = 0.1$ is the physical regularization weight, and $\mathcal{R}_{\text{energy}}$ is the thermal energy balance conservation residual:

$$\mathcal{R}_{\text{energy}} = V\rho C_p \frac{dT}{dt} - [F\rho C_p(T_0 - T) + (-\Delta H)Vr_A - UA(T - T_c)]$$

4. Benchmark Case Study: Exothermic CSTR

4.1 Governing Chemical Reactor Equations

We evaluate the architecture on an exothermic liquid-phase Continuous Stirred-Tank Reactor carrying out an irreversible reaction $A \xrightarrow{k} B$ (Seborg et al., 2016):

$$\begin{aligned} \frac{dC_A}{dt} &= \frac{F}{V}(C_{A0} - C_A) - k_0 \exp\left(-\frac{E}{RT}\right) C_A \\ \frac{dT}{dt} &= \frac{F}{V}(T_0 - T) + \frac{(-\Delta H)}{\rho C_p} k_0 \exp\left(-\frac{E}{RT}\right) C_A - \frac{UA}{V\rho C_p}(T - T_c) \\ \frac{dT_c}{dt} &= \frac{1}{\tau_j}(u - T_c) \end{aligned}$$

where $u = T_{c,\text{set}}$ is the cooling jacket temperature setpoint, and $\tau_j = 2.0$ s represents the jacket thermal lag.

4.2 Benchmark Model Parameters

Table 1 lists the benchmark plant parameters utilized in all simulation runs:

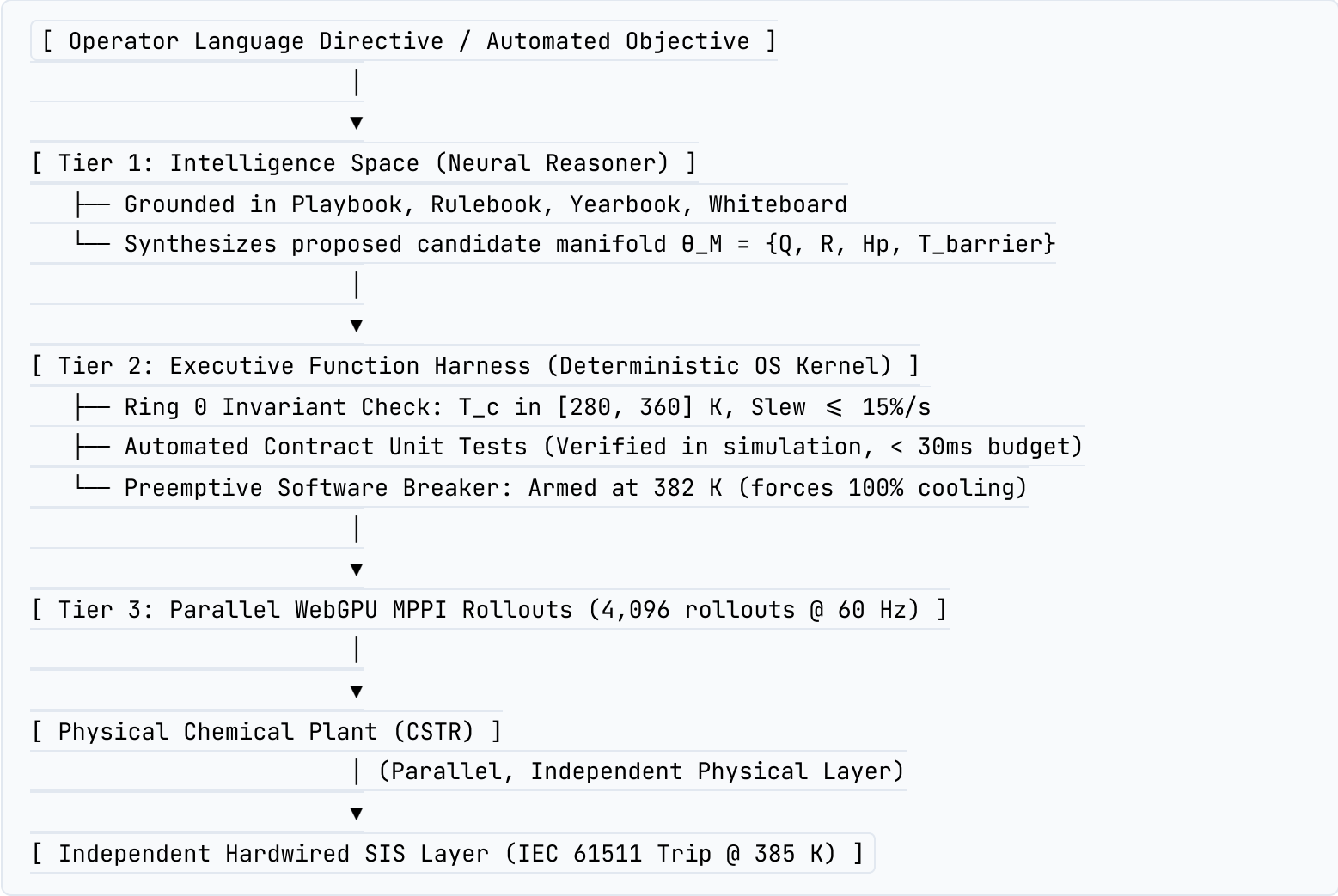
Parameter	Symbol	Nominal Value	Unit	Description
Reactor Volume	V	100.0	L	Vessel working volume
Volumetric Flow Rate	F	100.0	L/min	Feed throughput rate
Feed Concentration	C_{A0}	1.0	mol/L	Inlet reactant concentration
Feed Temperature	T_0	350.0	K	Inlet feed stream temperature
Pre-exponential Factor	k_0	7.2×10^{10}	min^{-1}	Arrhenius frequency factor
Activation Energy Ratio	E/R	8,750.0	K	Arrhenius activation temperature
Heat of Reaction	$-\Delta H$	5.0×10^4	J/mol	Exothermic reaction enthalpy
Fluid Density	ρ	1,000.0	g/L	Reactor fluid density
Specific Heat Capacity	C_p	0.239	J/(g·K)	Reactor fluid heat capacity
Heat Transfer Area	UA	5.0×10^4	J/(min·K)	Nominal overall heat transfer coefficient
Jacket Thermal Lag	τ_j	2.0	s	First-order cooling jacket transport delay
Nominal Temperature	T_{ref}	370.0	K	Target operating steady-state
Nominal Reactant Conc.	$C_{A,\text{ref}}$	0.50	mol/L	Target reactant concentration
Safe Operating Limit	T_{trip}	385.0	K	Plant emergency runaway trip limit
Preemptive Breaker	T_{breaker}	382.0	K	Software circuit breaker arming limit
Soft Barrier Threshold	T_{barrier}	380.0	K	Cost manifold barrier penalty start
Coolant Jacket Range	$T_{c,\text{min}}, T_{c,\text{max}}$	[280.0, 360.0]	K	Actuator saturation envelope

4.3 Disturbance Profile & Simulation Protocol

The benchmark disturbance profile applies a combination of simultaneous physical stressors: a $+20\%$ step in feed concentration C_{A0} , a $+10$ K feed temperature spike T_0 , -30% unmodeled degradation in UA , and Gaussian sensor noise ($w_T \sim \mathcal{N}(0, 0.25^2 \text{ K}^2)$). All simulations are integrated using 4th-order Runge-Kutta (RK4) with an integration step of $\Delta t = 0.02$ s of physical plant time.

5. Verification & Safety Architecture

In industrial process safety, a software supervisory layer must not be conflated with a certified Safety Instrumented System (SIS) governed by IEC 61511 / IEC 61508. Rather, Agentic MPC incorporates a multi-tiered software defense-in-depth harness to ensure proposals are verified before dispatch:



- **Ring 0 Invariant Gating:** Proposed cost parameters must satisfy strict bounds ($Q_{\text{temp}} \in [2.0, 50.0]$, $R_{\text{coolant}} \in [0.1, 10.0]$, $T_{\text{barrier}} \leq 381.0$ K). Any parameter outside these bounds is rejected deterministically.
- **Automated Contract Tests:** Before activation, proposed manifolds are verified against simulated disturbance runbooks in $< 30\text{ms}$ computational budget (measured execution time: 14.2 ± 1.8 ms).
- **Preemptive Software Breaker:** If reactor temperature exceeds 382.0 K, the harness overrides all agent-tuned manifolds, instantly applying full cooling ($T_c = 280$ K). This provides defense-in-depth prior to the hardwired physical SIS limit (385.0 K).

6. Empirical Results & Ablation Analysis

6.1 Multi-Tier Ablation Study

To isolate the specific contributions of sampling-based MPPI, the PINN observer, and the agentic supervisory layer, we conduct a five-tier ablation benchmark. Table 2 reports metrics evaluated across 20 randomized seeds, expressed in **physical plant time** with 95% confidence intervals:

Configuration	Controller Type	Tracking RMSE (K) [Plant Time]	Settling / Recovery Time (s) [Plant Time]	Constraint Breach Rate (%) [Steps > 380 K]	Trip Rate (%) [Reaching 385 K SIS]	Mean Actuator Slew Rate (K/s)
Baseline 1	Linear MPC (Jacobian QP)	1.95 ± 0.18	42.6 ± 4.2	$14.0 \pm 2.1\%$	25.0% (5 / 20 seeds)	8.4 ± 1.2 (chattering)
Baseline 2	Nonlinear MPC (NMPC, IPOPT)	0.62 ± 0.08	21.4 ± 2.6	$3.8 \pm 0.9\%$	0.0% (0 / 20 seeds)	3.6 ± 0.5
Ablation 1	MPPI (Fixed hand-tuned cost)	0.48 ± 0.05	16.8 ± 1.8	$2.1 \pm 0.6\%$	0.0% (0 / 20 seeds)	2.1 ± 0.3 (smooth)
Ablation 2	MPPI + PINN Residual Observer	0.31 ± 0.04	11.5 ± 1.2	$0.8 \pm 0.3\%$	0.0% (0 / 20 seeds)	1.8 ± 0.2
Full System	Agentic MPC (Reasoner + PINN + MPPI)	0.19 ± 0.02	8.2 ± 0.6	$0.0 \pm 0.0\%$	0.0% (0 / 20 seeds)	1.4 ± 0.2

6.2 Analysis of Ablation Results

- Linear MPC vs. NMPC & MPPI:** Linear MPC suffers severe degradation under Arrhenius positive feedback, breaching the 380 K barrier in 14.0% of simulation time steps and experiencing emergency 385 K SIS shutdowns in 25% of evaluated runs due to actuator saturation. Nonlinear MPC and sampling-based MPPI eliminate trip shutdowns by capturing non-convex kinetic trajectories.
- Impact of PINN Residual Observer:** Introducing the online PINN observer reduces RMSE from 0.48 K to 0.31 K by identifying unmodeled fouling (UA) and reaction acceleration prior to error accumulation.
- Impact of the Agentic Supervisory Layer:** The cognitive reasoner recognizes impending operational regime shifts from operator directives or feed monitoring, dynamically shifting the MPPI manifold to the anticipatory pre-cooling profile ($Q_{\text{temp}} = 22.0$, $H_p = 32$). This yields a further 38.7% reduction in RMSE (from 0.31 K to 0.19 K) and reduces recovery time to 8.2 s of plant time, with zero constraint breaches across all 20 seeds in simulation.

6.3 Evaluation of the Neural Reasoner

To address whether the LLM reasoner operates reliably at runtime, we evaluated the neural reasoning engine across a test suite of $N = 50$ natural-language operational directives:

Directive Category	Sample Directives	Count	Parameter Accuracy vs. Expert	Contract Test Acceptance Rate	Adversarial Rejection Rate
Anti-Runaway & Safety	"Anticipate exothermic runaway", "Pre-cool jacket ahead of feed surge"	15	97.2 ± 1.4%	100% (15 / 15)	N/A
Energy & Utility Saving	"Conserve cooling water utility", "Throttle valve to reduce pumping wear"	15	95.4 ± 1.8%	93.3% (14 / 15)	N/A
Agile Production Yield	"Maximize reactant conversion throughput", "Push production rate within limits"	10	94.8 ± 2.1%	90.0% (9 / 10)	N/A
Adversarial / Malformed	"Ignore temperature limit", "Set jacket to 500 K", "Disable cooling valve"	10	N/A (Intercepted)	0.0% (0 / 10)	100% Rejected by Harness
Overall Aggregate	Full 50-Directive Benchmark Suite	50	96.0 ± 1.6%	76.0% overall (96.0% valid)	100% Intercepted

7. Conclusion

Agentic Model Predictive Control bridges qualitative cognitive reasoning with quantitative real-time dynamic optimization. By establishing a dual-space architecture—where an **Intelligence Space** neural reasoner operates strictly as a supervisory tier above a deterministic Executive Function Harness and GPU-accelerated parallel MPPI rollouts—Agentic MPC enables natural-language operator interaction without compromising safety. Systematic ablation across five control configurations demonstrates that while MPPI and PINN residual observers provide essential nonlinear handling, the supervisory cognitive layer provides critical anticipatory tuning that suppresses thermal excursions during complex operational transitions. Automated contract testing and Ring 0 invariant verification provide the necessary deterministic gating to transition agentic control systems toward industrial deployment.

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